

Bangabandhu Sheikh Mujibur Rahman University, Kishoreganj
Department of Computer Science and Engineering
2nd Year 1st Semester B. Sc. (Engg.) In-course Examination 2023

Course Code: CSE 2101
Exam Duration: 60 Minutes

Course Title: Object Oriented Programming
Full Marks: 25

Answer all the questions

1. Why main method in Java is declared as static? Write down the difference between method overloading and method overriding with an example in Java. 5
2. What are the rules for defining the constructor? Describe constructor overloading with an example in Java. How are constructors called in the case of inheritance? 5
3. "Method overloading is not possible by changing the return type of the method only"- **Justify the statement.** 5

Using inheritance, one class can acquire the properties of others. Consider the following Animal class:

```
class Animal{  
    void walk(){  
        System.out.println("I am walking");  
    }  
}
```

This class has only one method, walk. Next, we want to create a Bird class that also has a fly method. We do this using extends keyword:

```
class Bird extends Animal {  
    void fly() {  
        System.out.println("I am flying");  
    }  
}
```

Finally, we can create a Bird object that can both fly and walk.

```
public class Solution{  
    public static void main(String[] args){  
        Bird bird = new Bird();  
        bird.walk();  
        bird.fly();  
    }  
}
```

The above code will print:

```
I am walking  
I am flying
```

This means that a Bird object has all the properties that an Animal object has, as well as some additional unique properties. **The code above is provided for you. You must add a sing method to the Bird class, then modify the main method accordingly so that the code prints the following lines:**

```
I am walking  
I am flying  
I am singing
```

4. How to achieve the encapsulation? Describe it with an example in Java. Mention a real-life example of encapsulation. 5
5. Describe the use of the following keyword with example in Java. 5
 - a. this
 - b. super